



Direct photo-oxidation of methane to methanol over a mono-iron hydroxyl site

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Natural gas, consisting mainly of methane (CH₄), has a relatively low energy density at ambient conditions (~36 kJ l⁻¹). Partial oxidation of CH₄ to methanol (CH₃OH) lifts the energy density to ~17 MJ l⁻¹ and drives the production of numerous chemicals. In nature, this is achieved by methane monooxygenase with di-iron sites, which is extremely challenging to mimic in artificial systems due to the high dissociation energy of the C–H bond in CH₄ (439 kJ mol⁻¹) and facile over-oxidation of CH₃OH to CO and CO₂. Here we report the direct photo-oxidation of CH₄ over mono-iron hydroxyl sites immobilized within a metal–organic framework, PMOF-RuFe(OH). Under ambient and flow conditions in the presence of H₂O and O₂, CH₄ is converted to CH₃OH with 100% selectivity and a time yield of 8.81 ± 0.34 mmol g_{cat}⁻¹ h⁻¹ (versus 5.05 mmol g_{cat}⁻¹ h⁻¹ for methane monooxygenase). By using operando spectroscopic and modelling techniques, we find that confined mono-iron hydroxyl sites bind CH₄ by forming an [Fe–OH...CH₄] intermediate, thus lowering the barrier for C–H bond activation. The confinement of mono-iron hydroxyl sites in a porous matrix demonstrates a strategy for C–H bond activation in CH₄ to drive the direct photosynthesis of CH₃OH.

The activation of the C–H bond in methane (CH₄) is a major goal in catalysis^{1–3}. Homogeneous catalysts based upon transition metal complexes and polyoxometalates can activate C–H bonds in CH₄. These systems rely heavily on highly corrosive reagents, such as sulfuric acid⁴ and trifluoroacetic acid⁵, to stabilize the reacted CH₄ and thus have substantial environmental impacts. Moreover, high pressure (usually >20 bar) is required for sufficient CH₄ to dissolve in solution. Inspired by methane monooxygenase in nature, iron- or copper-exchanged zeolites have been applied to promote the activation of the C–H bond in CH₄ (refs. 6–9). However, these processes operate at high temperature (>800 °C) and/or high pressure (>30 bar) and are thus highly energy intensive. Although the use of hydrogen peroxide (H₂O₂) can drive the desired reaction under milder conditions^{10,11}, its high cost prohibits large-scale application. On the other hand, O₂ is a cheap, readily available oxidant, but often leads to over-oxidation of CH₄ to CO₂ as the main product¹².

Photocatalysts based on BiVO₄, WO₃ or supported noble metals (for example, Au, Ag, Pt, Pd) on TiO₂ and ZnO can promote CH₄ activation at room temperature^{13–18}. Recently, photo-oxidation of CH₄ to CH₃OH has been achieved over FeO_x/TiO₂ using H₂O₂ as oxidant with a CH₃OH selectivity of 90% and a time yield of 0.36 mmol g_{cat}⁻¹ h⁻¹ (ref. 17). Porous metal–organic framework (MOF) materials incorporating immobilized catalytically active centres can act as thermocatalysts for CH₄ activation, although high temperature and pressure remain critical to promote the activation of C–H bonds^{19–21}. To date, light-driven partial oxidation of CH₄ over MOF catalysts has not been achieved. Here we describe the efficient direct photo-oxidation of CH₄ to CH₃OH under ambient conditions over

mono-iron hydroxyl sites confined in a MOF using H₂O and O₂ as oxidants. By judicious design of our catalytic system, an exceptional Fe turnover number (up to 2,269 ± 88), catalytic stability (>120 h) and an unprecedented time yield of 8.81 ± 0.34 mmol g_{cat}⁻¹ h⁻¹ (or 18,908 ± 730 mmol mol_{Fe}⁻¹ h⁻¹) for the production of CH₃OH from CH₄ can be achieved.

Synthesis and characterization

Photocatalytic systems integrating multiple components can underpin synergistically the observed catalytic performance. UiO-67 was chosen as a functional platform to integrate the three crucial components of our photocatalyst: the photosensitizer (PS) [Ru^{II}(bpy)₃(bpydc)] (L), (bpy: 2,2′-bipyridine; H₂bpydc: 2,2′-bipyridine-5,5′-dicarboxylic acid) to absorb light; a multi-electron redox-active polyvanadotungstate [PW₉V₃O₄₀]⁶⁻ (PW₉V₃) to activate O₂; a mono-iron hydroxyl site to activate CH₄ (Fig. 1a and Supplementary Fig. 1). The PS and PW₉V₃ were incorporated into porous UiO-67 in a one-pot synthesis to give PMOF-Ru, and the iron sites were then introduced by post-synthetic metalation with FeCl₃•6H₂O to afford PMOF-RuFe(Cl), ((PW₉V₃)_{0.1}@Zr₆(μ₃-O)_{6.1}(μ₃-OH)_{1.9}(H₂O)_{0.24}(bpydc)_{3.7}[(bpydc)FeCl₂(THF)₂]_{1.9}(L)_{0.28}). The structure of PMOF-RuFe(Cl) was established by a combination of thermogravimetric and elemental analyses, by NMR, infrared (IR) and XAS (X-ray absorption) spectroscopies, and by consideration of charge balance (Supplementary Figs. 2–4 and Supplementary Table 1). Powder X-ray diffraction patterns and scanning electron microscopy (SEM) images confirm the retention of UiO-type structure and morphology in PMOF-RuFe(Cl) (Fig. 2a and Supplementary Fig. 5), and binding of Fe^{III} to bpydc²⁻ was

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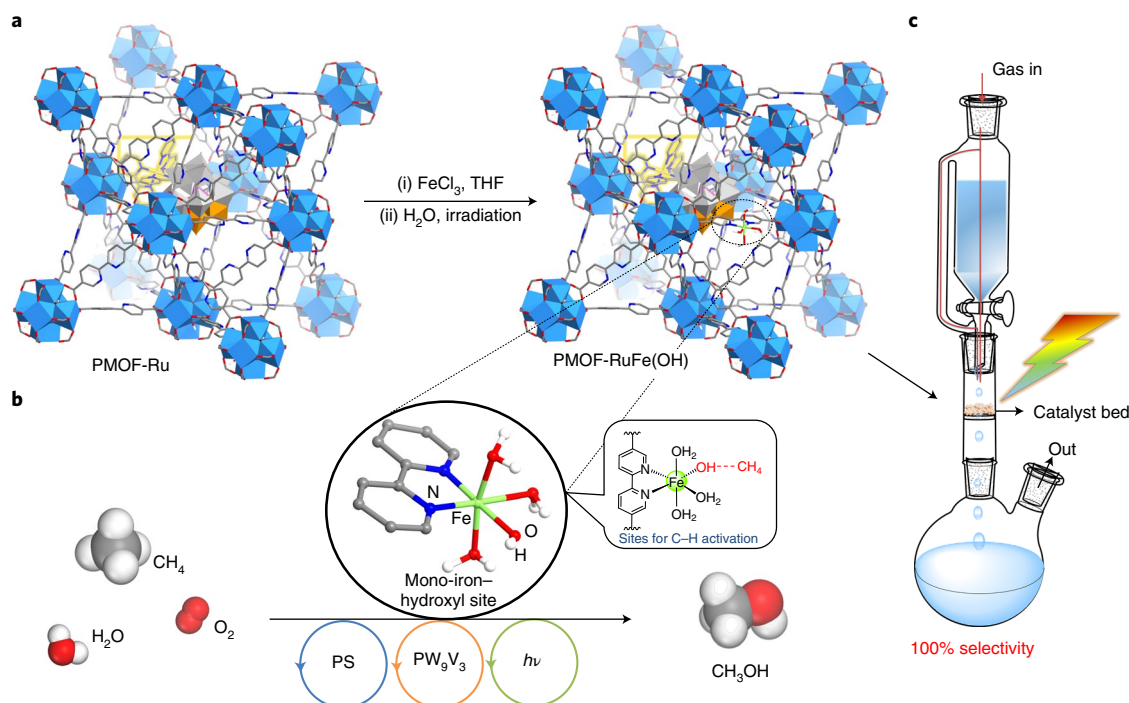


Fig. 1 | Design and synthesis of the PMOF-RuFe(OH) catalyst and the flow reactor for photo-oxidation of CH₄ to CH₃OH. **a**, View of the structure of PMOF-RuFe(OH) incorporating mono-iron hydroxyl, [Ru^{II}(bpy)₂(bpydc)] and [PW₉V₃O₄₀]⁶⁻ species within the pore of UiO-67. All three components are shown in one unit cell to simplify the overall model, and a homogeneous distribution of these components across the entire structure is observed (Supplementary Fig. 25). The structural model has been determined by INS analysis and DFT calculations, coupled with powder X-ray diffraction, NMR, IR and XAS analysis, and consideration of charge balance. Labelling scheme: C, grey; O, red; H, white; N, blue; Fe, green; V, orange; W, dark grey; P, purple; {Zr₆} cluster, blue polyhedral; [Ru^{II}(bpy)₂(bpydc)] (PS), gold. **b**, Schematic showing synergistic catalysis in PMOF-RuFe(OH) for selective photo-activation of the C-H bond in CH₄ to CH₃OH in the presence of oxygen and water. *hν*, irradiation. **c**, View of the continuous-flow apparatus for photocatalytic conversion of CH₄ to CH₃OH over PMOF-RuFe(OH). The wavelength of light used is 400–780 nm.

supported by a pronounced absorption band centred at 370 nm in the UV-visible (UV-vis) spectra due to metal-to-ligand charge transfer (Fig. 2b). The homogeneous distribution of Ru, Fe and W throughout PMOF-RuFe(Cl) has been confirmed by energy dispersive X-ray analysis (Supplementary Table 2). The surface area of PMOF-Ru is reduced by ~50% on post-synthetic metalation to form PMOF-RuFe(Cl) (Supplementary Fig. 6 and Supplementary Table 3).

On pre-treatment of PMOF-RuFe(Cl) in water with light for 2 h, an active catalyst PMOF-RuFe(OH) is generated, in agreement with an induction period for CH₃OH formation catalysed by PMOF-RuFe(Cl) (Supplementary Fig. 7). The structure of PMOF-RuFe(OH) was established by inelastic neutron scattering (INS) and density functional theory (DFT) calculations (Fig. 1 and Supplementary Fig. 1b,c). The chemical states of the active species were interrogated by X-ray photoelectron spectroscopy analysis of PMOF-RuFe(OH), which shows typical Fe^{III} signals, with the peaks at 714 and 727 eV related to hydroxyl groups, indicating the presence of Fe-OH species, as in FeO(OH) (Supplementary Fig. 8)^{17,22}. Raman spectra confirmed the absence of O-O bonds, thus excluding the formation of Fe-O₂ adducts or Fe-OOH species (Supplementary Fig. 9)²³. Electron paramagnetic resonance (EPR) spectra of PMOF-RuFe(Cl) and PMOF-RuFe(OH) at low temperature (5 K) are characteristic of high-spin Fe^{III} states with highly rhombic environments, with signals observed at *g* ≈ 4.3 (Supplementary Fig. 10 and Supplementary Table 4)²⁴. X-ray absorption near-edge structure (XANES) analysis of PMOF-RuFe(OH) shows the iron K edge energy of 7,114.8 eV, which corresponds to the transition of 1s → 3d and shows a slight blue shift and reduced strength

compared with PMOF-RuFe(Cl) (7,114.4 eV), FeCl₃ and Fe₂O₃ (Fig. 2c)²⁵. The extended X-ray absorption fine structure (EXAFS) spectrum of PMOF-RuFe(OH) is entirely consistent with a distorted octahedral complex, [(bpy)Fe(OH)(H₂O)₃]²⁺, which has been optimized by DFT calculations (Fe-N_{bpy} = 2.16 ± 0.10, 2.10 ± 0.10 Å; Fe-O_{OH} = 1.78 ± 0.09 Å; Fe-O_{H₂O} = 2.17 ± 0.09 Å (×3)) (Fig. 2d, Supplementary Table 5, Supplementary Fig. 11 and Supplementary Text, *pK_a* calculations). Analysis of the EXAFS and wavelet spectra confirms the absence of a secondary correlation from the scattering of binuclear or cluster {Fe...Fe} species in *R*-space; the Fe...Fe distances in Fe₂O₃ and iron nanoparticles are ~2.5 and 2.2 Å, respectively (Fig. 2e,f and Supplementary Figs. 12 and 13). This is entirely consistent with the ⁵⁷Fe Mössbauer transmission and XANES analysis (Supplementary Figs. 14 and 15). In addition, the V centres in PW₉V₃ are partly reduced to V^{IV} without structural change (Supplementary Figs. 8b and 16 and Supplementary Table 6).

Photocatalytic performance in batch mode

The photocatalytic performance of these catalysts was investigated first in batch mode in water with CH₄/O₂ (v/v = 1/1, 1 atm) as feed gas and irradiation with visible light (λ = 400–780 nm) at 25 °C (Supplementary Fig. 17). PMOF-RuFe(OH) exhibits an exceptional CH₃OH productivity of 629 ± 68 μmol as the only liquid product after 20 h of illumination, with the time yield of CH₃OH being 3,145 ± 340 μmol g_{cat}⁻¹ h⁻¹ (or 6,750 ± 730 mmol mol_{Fe}⁻¹ h⁻¹) and a cumulative Fe turnover number (Fe-TON) of 135 ± 16 (Supplementary Fig. 18 and Supplementary Table 7). This outperforms all state-of-the-art photocatalysts operating at ambient conditions (Fig. 3a and Supplementary Table 8)^{26–37} and

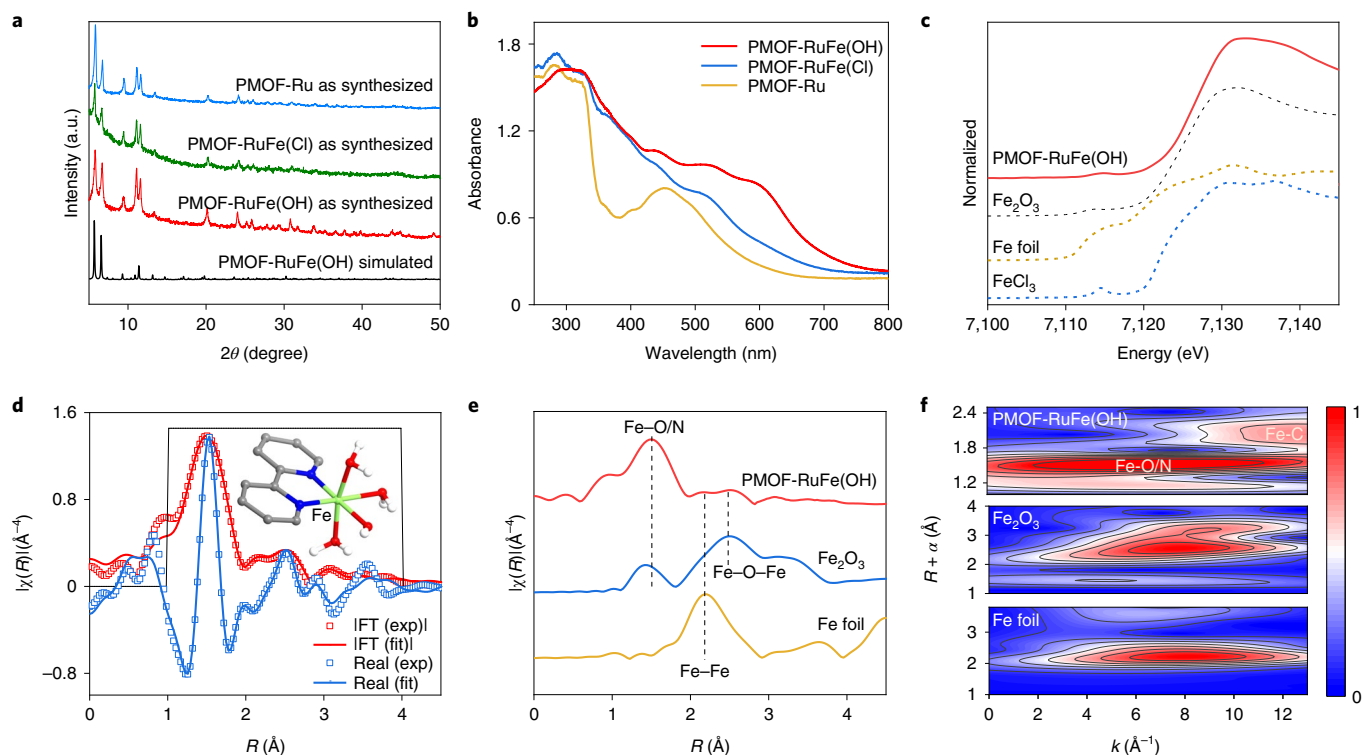


Fig. 2 | Characterization of PMOF-RuFe(OH). **a**, Powder X-ray diffraction patterns of PMOF-RuFe(OH), PMOF-Ru and PMOF-RuFe(Cl), and the simulated pattern. All materials retain the UiO-67 framework structure. **b**, Diffuse reflectance UV-vis-NIR spectra. The band at ~465 nm corresponds to the PS, and the absorption at ~590 nm is attributed to the Fe-OH group. **c**, Normalized Fe K edge XANES spectra of PMOF-RuFe(OH), Fe₂O₃, Fe foil and FeCl₃. **d**, EXAFS spectra and fitted spectra using the model of [(bpy)Fe(OH)(H₂O)₃]²⁺ (inset: Fe, green; O, red; N, blue; C, grey and H, white) in *R*-space (1–4 Å, black lines) at the Fe K edge of PMOF-RuFe(OH) with the magnitude (hollow squares, red) and real components (hollow squares, blue) of the Fourier transforms (FT). exp, experiment; fit, fitted. **e**, *k*²-weighted Fe K edge EXAFS spectra (**e**) and wavelet transform of PMOF-RuFe(OH), Fe₂O₃ and Fe foil (**f**).

most of the thermocatalysts, including many systems that operate under high-temperature and/or high-pressure conditions (Supplementary Table 9)^{38,39}. Recently, an Au-Pd/TiO₂ catalyst has shown a high CH₃OH yield of 12.6 mmol g_{cat}⁻¹ h⁻¹ with a selectivity of 42.3% for the photoconversion of CH₄ at 40 bar (ref. 40). No gaseous (<0.1 ppm) product, except a trace amount of CO₂ (11.4 ± 0.3 μmol), was detected for catalysis with PMOF-RuFe(OH) (Supplementary Fig. 19). A selectivity of 98% to CH₃OH was achieved over PMOF-RuFe(OH) under ambient conditions using H₂O and O₂ as oxidants, with a concentration of CH₃OH of ~210 mmol l⁻¹ in the resultant solution. Reaction using ¹³C-labelled ¹³CH₄ yielded ¹³CH₃OH as the only product (Supplementary Fig. 20), and the experiment performed in the absence of CH₄ gave no carbon-containing product (Supplementary Table 7, entry 4). Thus, CH₄ is the sole carbon source in these experiments. Control experiments: (1) without any catalysts; (2) without light; and (3) without light but with heating at 100 °C: all yielded negligible products (Supplementary Table 7, entries 1, 5, 6), confirming the photo-driven nature of the reaction.

The effect of H₂O and O₂ in this reaction was explored. In the absence of H₂O or using anhydrous benzene as solvent, CO₂ was the only product and no liquid products were detected (Supplementary Fig. 21 and Supplementary Table 10). In the presence of H₂O, both the selectivity and production of CH₃OH increased, and formation of CO₂ remained at <2% over the 20-hour period, confirming low levels of over-oxidation of CH₃OH under these conditions, consistent with the reported water-driven oxidation of CH₄ (refs. 14,36,39). Experiments with varying levels of O₂ were also conducted, with formation of CH₃OH decreasing dramatically in the absence of O₂ (Supplementary Table 11). Interestingly, CH₃OH production is

observed to be similar using air or O₂ (607 versus 629 μmol yield, respectively) under the same conditions. This reflects the low solubility and the barrier to activation of CH₄, which renders the overall rate of reaction more sensitive to the pressure of CH₄ than O₂, as confirmed by dependence studies (Supplementary Figs. 21 and 22). The isotopic labelling experiment using H₂¹⁸O/¹⁶O₂ further validated the participation of both, to give 75% of CH₃¹⁶OH and 25% of CH₃¹⁸OH (Supplementary Fig. 23).

The reusability of PMOF-RuFe(OH) was confirmed by the consistent activity over ten consecutive cycles, adding up to a total reaction time of 200 h (Supplementary Fig. 24a). This is in contrast with most state-of-the-art semiconductor-based photocatalysts, which have limited durability (typically <10 h)^{18,33,41}. SEM, transmission electron microscopy, powder X-ray diffraction, IR, UV-vis, N₂ isotherms and XAS studies of the used catalyst show little change in morphology (Supplementary Figs. 24–28) and oxidation state compared with the fresh catalyst (Supplementary Figs. 8 and 29), as well as negligible leaching of active sites. Actinometric measurements show a high quantum efficiency of 0.07 for PMOF-RuFe(OH).

The catalytic performance of materials missing one or two of the three integrated components was investigated, as were physical mixtures of these various components. Thus, MOF-Ru (comprising UiO-67 with the PS [Ru^{II}(bpy)₂(bpydc)] but without PW₅V₃), MOF-RuFe(Cl) and MOF-RuFe(OH) were prepared, characterized and investigated. In all cases only trace amounts of CH₃OH were obtained (Fig. 3b). Fe₂O₃ nanoparticles and Fe₂O₃/PMOF-Ru were also examined but gave no product, thus ruling out possible formation of Fe₂O₃ nanoparticles as active species (Supplementary Table 7, entries 15 and 16). This implies that it is the positional anchoring of all three components and their synergic cooperation

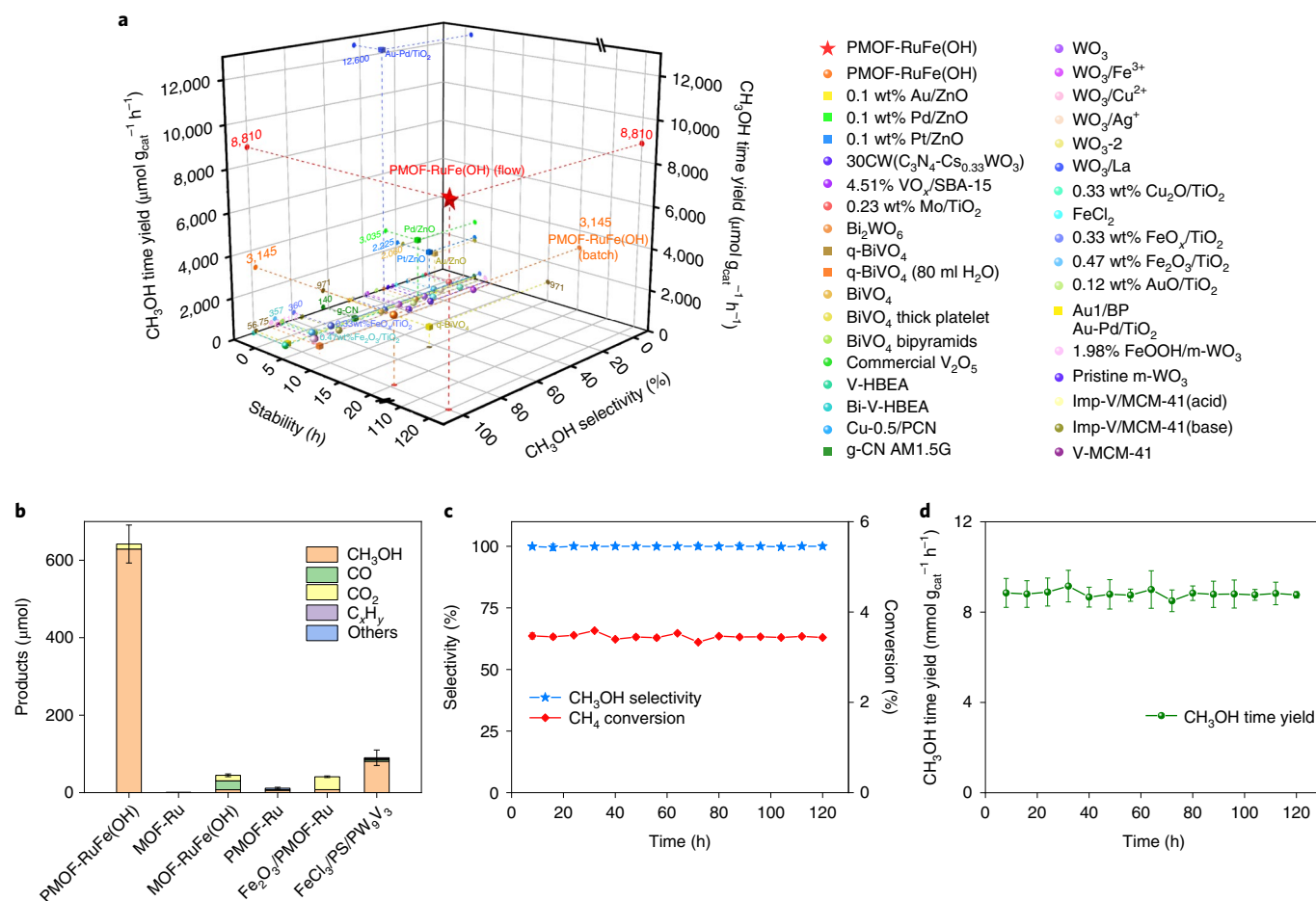


Fig. 3 | Photocatalytic performance. **a**, Comparison of the catalytic activity for photo-oxidation of CH₄ to CH₃OH over PMOF-RuFe(OH) with other photocatalysts^{15,18,26–37,40}. Red stars and orange spheres represent the activity of PMOF-RuFe(OH) in flow and batch mode, respectively. Other spheres represent the activity obtained at ambient pressures, and the cubes represent the activity obtained at high pressures. The time yield for CH₃OH production is based on the amount of CH₄ converted to CH₃OH per hour per weight of catalyst; full data are listed in Supplementary Table 8. **b**, Comparison of the catalytic activity over various catalysts in batch mode. Reaction conditions: 3 ml water, 20 h visible light, 10 mg catalyst and CH₄/O₂ (1 atm). **c,d**, CH₃OH selectivity and CH₄ conversion (**c**) and time yield (**d**) for formation of CH₃OH over a period of 120 h on stream over PMOF-RuFe(OH) in flow mode. Error bars give the standard deviation from three independent tests.

within PMOF-RuFe(OH) that results in the observed photocatalytic activity.

Mono-iron hydroxyl sites for CH₄ activation

We next sought to identify the binding sites for CH₄ molecules. Isotherms for CH₄ uptake in both PMOF-RuFe(OH) and MOF-RuFe(OH) (that is, in the absence of PW₉V₃) display strong adsorption of CH₄, as evidenced by the initial steep rise and marked hysteric desorption (residue of 0.5 CH₄/Fe). This behaviour is not observed in PMOF-Ru (Fig. 4a), suggesting that the mono-Fe^{III} species acts as a binding site for CH₄. INS spectra coupled with DFT calculations enabled the direct visualization of the binding dynamics of CH₄ in this system (Fig. 4b, Supplementary Text and Supplementary Figs. 30–32). At low loadings of CH₄ in PMOF-RuFe(OH) (0.22 and 0.45 CH₄/Fe), the intensity at <40 meV increases due to the libration/vibration of adsorbed CH₄ molecules and their recoil. The difference spectra of bare and CH₄-loaded PMOF-RuFe(OH) reveal three major peaks. The broad peak I centred at 12 meV corresponds to the restricted/hindered motion of CH₄ on the adsorption site and is reminiscent of the libration/vibration in solid CH₄ (ref. 42). Peaks II (50–55 meV) and III (65–70 meV) are assigned to changes in INS intensities of the MOF on adsorption of CH₄, specifically

in the (i) vibration of the –OH group with H moving perpendicular and parallel to the [(bpy)Fe–O_{OH}] plane, (ii) wagging modes of H₂O located out of and in the [(bpy)Fe–O_{OH}] plane, (iii) wagging modes of H₂O located in the [(bpy)Fe–O_{OH}] plane and vibration of OH with H moving perpendicular to O–H within the [(bpy)Fe–O_{OH}] plane and (iv) rocking modes of the two axial H₂O ligands in the [(bpy)Fe(OH)(H₂O)₃]²⁺ fragment (Fig. 4b, Supplementary Fig. 32). At a higher loading of 1.1 CH₄/Fe, a notable increase in intensity of peak I is observed, but not of peaks II or III, indicating that the additional CH₄ has a much weaker or zero interaction with the MOF structure. Interestingly, peaks I–III are all retained on evacuating the CH₄-loaded PMOF-RuFe(OH) sample under dynamic vacuum at ambient temperature, which is consistent with retention and strong binding of CH₄ within the pores. DFT calculations show that the interaction between [(bpy)Fe(OH)(H₂O)₃]²⁺ and CH₄ in PMOF-RuFe(OH) is primarily based on hydrogen bonds (Fe–OH...CH₄ = 2.39 Å) (Supplementary Fig. 33). This is also confirmed by a red shift of 3 cm^{–1} for the C–H stretching vibration for adsorbed CH₄ in PMOF-RuFe(OH), measured by in situ synchrotron Fourier transform-IR (FTIR) spectroscopy (Supplementary Fig. 34)⁴³. In contrast, no shift in the C–H stretching region in the FTIR spectrum is observed for CH₄-loaded PMOF-RuFe(Cl), which

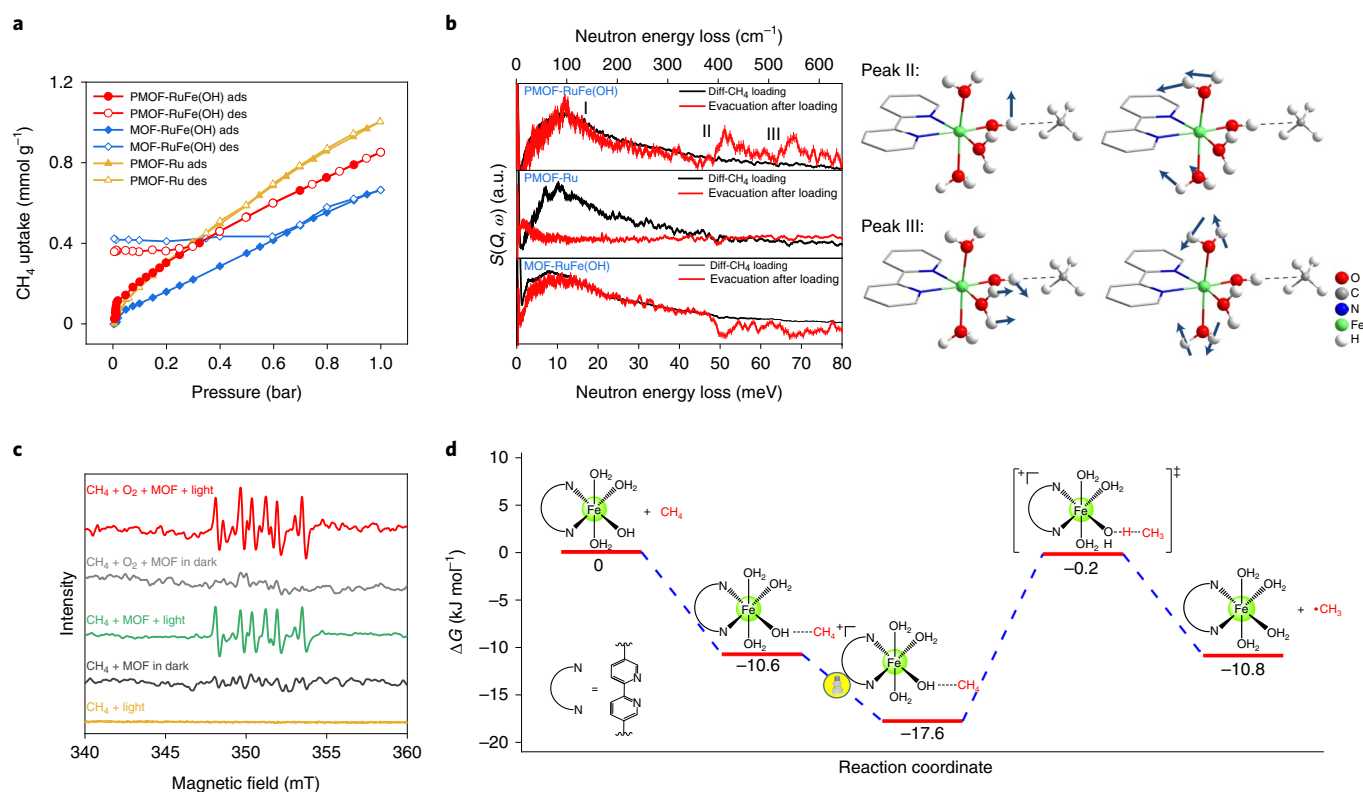


Fig. 4 | Investigation of the mono-iron hydroxyl site for C-H activation. **a**, Adsorption (ads)/desorption (des) isotherms for CH₄ in PMOF-RuFe(OH), PMOF-Ru and MOF-RuFe(OH) at 298 K. **b**, Comparison of the difference (diff) INS spectra of bare and CH₄-loaded sample and the difference INS spectra on evacuation for PMOF-RuFe(OH), PMOF-Ru and MOF-RuFe(OH). Vibrational modes of peaks II and III are shown on the right. For both PMOF-RuFe(OH) and MOF-RuFe(OH), there is a clear signal of residual CH₄ on desorption, whereas PMOF-Ru shows complete desorption. **c**, EPR spectra of PMOF-RuFe(OH) under different conditions. Minor signals for •DMPO-OH radical within the 'CH₄ + MOF + light' experiment were observed due to water splitting. **d**, Calculated energy profiles for the activation of C-H bond in CH₄ over confined mono-iron hydroxyl sites. The initial structure is [(bpy)Fe(OH)(H₂O)₃]²⁺ and an electron-transfer process occurs on the complex of [(bpy)Fe(OH)(H₂O)₃...CH₄]²⁺ to achieve activation of CH₄. No Fe(IV) species were detected in the in situ EPR experiment, suggesting a ligand-based redox process. The charges are determined on the basis of the natural population analysis and spin densities (Supplementary Fig. 38 and Supplementary Tables 13–16).

is consistent with the critical role of mono-iron hydroxyl sites in binding CH₄.

EPR spectroscopic spin-trapping experiments combined with DFT calculations afford further insights into the C-H activation process. The EPR spectrum of irradiated CH₄-loaded PMOF-RuFe(OH) with 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) in water shows a six-line signal typical of the DMPO-CH₃ radical adduct, suggesting the photo-induced formation of methyl radicals (Fig. 4c, Supplementary Fig. 35a and Supplementary Table 12). A minor component, DMPO-OH, was also observed, presumably due to splitting of water. This is consistent with the observation that the catalytic system can be quenched by radical scavengers (Supplementary Fig. 36). In contrast, no radical adduct was observed in the absence of catalysts or if the reaction was carried out over PMOF-Ru (Fig. 4c and Supplementary Fig. 35b). This suggests that CH₄ adsorbed at the iron-hydroxyl sites attains a lower energy state with a reduced activation barrier to the formation of •CH₃ radicals (Fig. 4d and Supplementary Text). Interestingly, DFT calculations suggest that the radical is not completely 'free' but is stabilized by the iron-hydroxyl sites to form Fe(OH₂)...CH₃ interactions [H₂O...CCH₃ = 2.02 Å], thus importantly inhibiting C-C coupling reactions (Supplementary Figs. 37 and 38 and Supplementary Tables 13–16). EPR trapping studies undertaken as a function of O₂ also show signals for •DMPO-OH (Fig. 4c and Supplementary Fig. 39), implying that •OH is the reactive oxygen intermediate

rather than ¹O₂, O₂^{•-} or •OOH; this is also demonstrated by the corresponding quenching experiments (Supplementary Table 17).

Mechanism studies

We propose that the photocatalytic cycle is initiated by photo-induced electron transfer from the excited-state PS' of [Ru(bpy)₂(bpydc)]⁺ to produce reduced PW₉V₃^{IV/V}, as confirmed by the in situ EPR spectroscopy (Supplementary Fig. 40, blue line), photoluminescent (Supplementary Figs. 41–43 and Supplementary Table 18) and electrochemical studies (Supplementary Fig. 44). The photocurrent response and electrochemical impedance spectroscopic studies confirm a notable enhanced efficiency for charge separation in PMOF-RuFe(OH) compared with partially functionalized systems (Supplementary Fig. 45). The electron-rich PW₉V₃^{IV/V} then drives the photoreduction of O₂ to H₂O₂ via proton-coupled (from H₂O) electron transfer (from reduced PW₉V₃^{IV/V}), and H₂O₂ can be detected after illumination of PMOF-Ru for 5 h in the presence of O₂. In contrast, no H₂O₂ was detected in the reaction over PMOF-RuFe(OH) owing to the rapid conversion of H₂O₂ formed in situ to •OH radicals promoted by the {Fe-OH} moiety under illumination^{44,45} (Supplementary Fig. 46), which is entirely consistent with the positive shift of the onset potential for reduction of H₂O₂ with PMOF-RuFe(OH) (Supplementary Fig. 47a). Also, addition of H₂O₂ enhances the production of CH₃OH under irradiation (Supplementary Table 17, entry 6). The formation of •OH radical

is also confirmed by a fluorescence spectroscopic study using terephthalic acid as a probe (Supplementary Fig. 47b). Thus, C–H activation proceeds via a pathway where the adsorbed CH_4 interacts with the mono-iron hydroxyl moiety and is oxidized readily to $\bullet\text{CH}_3$ radical, which is stabilized by the iron site (Supplementary Fig. 48). The H/D kinetic isotope effect (KIE, $k_{\text{H}}/k_{\text{D}}$) using CH_4 and CD_4 gave a value for KIE ($k_{\text{CH}_3\text{OH}}/k_{\text{CD}_3\text{OD}}$) of 3.0 ± 0.4 , which represents a typical value for the primary kinetic isotope effect and suggests that the C–H bond dissociation is the rate-determining step (Supplementary Figs. 49 and 50). CH_3OH is then formed by coupling of $\bullet\text{CH}_3$ and $\bullet\text{OH}$ radicals and extracted from the MOF by water. The high CH_3OH selectivity is attributed to its rapid desorption owing to the relatively weak adsorption within the MOF matrix in the presence of water (Supplementary Figs. 51–54). Also, light irradiation effectively quenches $\bullet\text{OOH}$ intermediates to avoid over-oxidation of CH_3OH (Supplementary Table 17, entries 7–10). To achieve optimal production of CH_3OH , all components (that is, MOF, photosensitizer, polyvanadotungstate and mono-iron hydroxyl species) within the cycle must function cooperatively. The UiO-67 framework provides a stable platform to achieve this synergy to afford a higher activity compared with soluble methane monooxygenase incorporating di-iron centres (Supplementary Fig. 55)⁷.

Continuous-flow catalysis

Building on the observed high activity of PMOF-RuFe(OH), we have developed a continuous-flow photo-oxidation process, where a flow of CH_4/O_2 -saturated water (15 ml h^{-1}) is passed through a packed layer of PMOF-RuFe(OH) catalyst under irradiation (Fig. 1c). Importantly, the reacted CH_4 over PMOF-RuFe(OH) includes both the water-dissolved CH_4 and part of the gaseous CH_4 passing through the branch of the dropping funnel (marked with a red line in Fig. 1c). It is the latter that breaks the limitation of CH_4 solubility because CH_4 can be directly adsorbed by the MOF, a feature that is distinct from conventional non-porous photocatalysts. Thus, this creates a dynamic gas/solid/liquid interface to maximize the contact between CH_4 , O_2 , H_2O and the catalyst. Using this simple set-up, CH_3OH was obtained as the only product in water with 100% selectivity and an accumulated Fe-TON of $2,269 \pm 88$ with high catalyst stability over 120 h (Fig. 3c). This allows an unprecedented CH_3OH time yield of $8.81 \pm 0.34 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$ (or $18,908 \pm 730 \text{ mmol mol}_{\text{Fe}}^{-1} \text{ h}^{-1}$) to be achieved under entirely ambient conditions. PMOF-RuFe(OH) outperforms methane monooxygenase (*Methylococcus capsulatus* (Bath), $5.05 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$) and state-of-the-art photocatalysts, such as q-BiVO₄ ($0.97 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$), Pd/ZnO ($3.04 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$) and FeOOH/m-WO₃ ($0.21 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$)^{7,14,18,34}. An economic analysis of the synthesis of CH_3OH via this direct photochemical approach using CH_4 and via the state-of-the-art industrial process using syngas ($\text{CO}/\text{CO}_2/\text{H}_2$) over commercial Cu/ZnO/Al₂O₃ catalyst has been performed, demonstrating the great promise of PMOF-RuFe(OH) in practical applications (Supplementary Notes and Supplementary Fig. 56).

Outlook

CH_4 is a very plentiful carbon-based energy source, but it has a substantial environmental impact as a greenhouse gas. Its direct photoconversion to CH_3OH , a liquid fuel and carbon feedstock, is an important alternative route to harness this carbon feedstock. However, this is a highly challenging process to achieve since over-oxidation to CO_2 is commonly observed. The confinement of mono-iron hydroxyl sites within a porous metal–organic framework matrix, PMOF-RuFe(OH), linked to a photoactive receptor, represents an effective strategy for the activation of C–H bonds in CH_4 to drive the direct photosynthesis of CH_3OH without the use of H_2O_2 . Future effort will be devoted to assessing practical photo-oxidation of CH_4 to CH_3OH with high catalytic stability and high turnover.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41563-022-01279-1>.

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Methods

Safety considerations. Although all the reactions were carried out at ambient temperature, mixtures of CH₄ and O₂ can be explosive. Therefore, all glassware and apparatus should be pressure proof with full appropriate containment and protective screening.

Catalytic preparation. The photosensitizer, [Ru(bpy)₂(H₂bpydc)]Cl₂ and [(*n*-C₄H₉)₄N]₆[α-1,2,3-PW₉V₃O₄₀] (TBA-PW₉V₃) were synthesized according to the literature methods^{46,47}. ZrCl₄ (10 mg, 43 μmol), H₂bpydc (9 mg, 37 μmol), [Ru(bpy)₂(H₂bpydc)]Cl₂ (2 mg, 4 μmol) and glacial acetic acid (82 μl, 1.44 mmol) in the presence of TBA-PW₉V₃ (2 mg) were dissolved in 1.5 ml DMF and sealed in a 5 ml vial. The vial was heated at 100 °C for 72 h, and, after cooling to room temperature, the solid product was collected and sequentially washed several times with DMF and THF and dried at 50 °C under vacuum overnight. An orange powder, PMOF-Ru, was obtained in a yield of 52%. Inductively coupled plasma-optical emission spectrometry (ICP-OES) analysis confirmed the Zr/Ru ratio of 21.4, indicating 0.28 Ru per {Zr₄} cluster, respectively.

For post-synthetic metalation, FeCl₃•6H₂O (60 mg) was dissolved in 10 ml THF and the solution added to a suspension of MOF (60 mg) in THF (10 ml). The resulting mixture was sonicated for 0.5 h. After being stirred at room temperature overnight, the solid sample, PMOF-RuFe(Cl), was collected by centrifugation and washed with THF 4–5 times to obtain a colourless supernatant. The dried sample was then treated under visible-light irradiation in water for 2 h, and PMOF-RuFe(OH) was generated. The Fe loading was determined by ICP-OES analysis to be 2.6 wt%, corresponding to a Fe/Zr ratio of 0.31, and 1.9 Fe per {Zr₄} cluster.

Photocatalytic oxidation of CH₄ in batch mode. Photocatalytic oxidation of CH₄ was carried out in a transparent reaction vessel with a magnetic stirrer. A 10 mg portion of catalyst and 3 ml H₂O were added into a 50 ml Schlenk tube equipped with a cross-shaped magnetic stirrer bar, and the tube was sealed and evacuated by three successive freeze–pump–thaw cycles. Two balloons filled with CH₄ and O₂ (v/v = 1/1, 1 atm) were incorporated into the system through the necks of the tube, and the system was placed 10 cm in front of a Xe lamp with a 400 nm cut-off filter (λ = 400–780 nm). The reaction was stirred at 1,000 rpm for 20 h, after which the gas in the headspace of the tube was analysed both by gas chromatography (GC) equipped with a thermal conductivity detector and a Bruker Matrix MG5 FTIR spectrometer (resolution, 0.5 cm^{−1}). 1,4-Dioxane (100 μl) was added to the liquid solution as internal standard for GC analysis (column: HP-5, flame ionization detector). The organic phase was extracted with ethyl acetate and analysed by GC combined with ¹H NMR spectroscopy and GC-MS; the aqueous phase was analysed by HPLC.

Photocatalytic oxidation of CH₄ in flow mode. The reaction was performed with a designed fixed-bed reactor. A 10.0 mg portion of solid catalyst was packed onto the surface of a glass frit and placed underneath a dropping funnel filled with water. After the system was evacuated and sealed, the gas mixture of 1:1 CH₄:O₂ was introduced into the reactor from the top (2 ml min^{−1}). The rate of flow of water (~15 ml h^{−1}) through the catalyst bed was controlled by a valve, and the system was irradiated using a Xe lamp (λ = 400–780 nm) to promote reaction. The product was collected via the outlet into a round-bottomed flask and analysed for each 8 h run.

Inelastic neutron scattering. The INS scattering experiments were performed according to our reported method⁴⁸ at Spallation Neutron Source, Oak Ridge National Laboratory, USA. The loading of CH₄ into the cylindrical sample container was performed volumetrically at room temperature, and the temperature was reduced to 5 K for data collection. After reaching adsorption saturation, the cell was evacuated and cooled for INS scattering to detect the presence of strongly bound CH₄. The detailed procedure for the experiment is summarized in Supplementary Fig. 30. Background spectra for the bare MOF were subtracted to obtain the difference spectra. The modelling of the INS spectra is shown in Supplementary Methods.

Electron paramagnetic resonance. Continuous wave (CW) EPR measurements were carried out at X-band (~9.5 GHz) on a Bruker Elexsys 500 spectrometer equipped with an Oxford Instruments temperature control system. EPR data were collected with modulation amplitude 10 G, microwave frequency ~9.85 GHz and microwave power ~6.325 mW. The magnetic field was calibrated using 2-diphenyl-1-picrylhydrazyl with g = 2.0028. Theoretical modelling of EPR spectra was performed with the EasySpin (v5.2.24) package⁴⁹ embedded in MATLAB software v.2013a. Spin-trapping experiments were carried out at X-band (9.85 GHz) and room temperature using DMPO as the spin trap, and a commercial Bruker EMX Micro EPR spectrometer. A modulation amplitude of 2 G was used with

a microwave power of ~2.0 mW after testing for saturation of spectral lines. EPR measurements of the MOF catalysts were carried out as follows: the MOF sample was prepared by loading 25 mg into a J. Young quartz tube (outer diameter, 4 mm; inner diameter, 2.8 mm). The sample was activated under dynamic vacuum (<10^{−5} mbar) for 12 h at 80 °C. Measurements were then conducted on samples as a function of irradiation and various gas loadings of CH₄, O₂ and CH₄ + O₂.

DMPO was used as the spin-trapping reagent in these experiments. DMPO is a nitron spin trap, which is largely used for trapping short-lived radicals with the generation of a more stable free radical, which is readily detected by EPR spectroscopy. DMPO has a β-hydrogen and yields adduct with hyperfine splitting due to both the β-hydrogen and the nitroxide nitrogen⁵⁰. DMPO was dissolved in deoxygenated water and then added into the photocatalytic system in the presence of CH₄, O₂ or CH₄ + O₂. All the reagents and tubes were deoxygenated using Ar before use, and 0.5 ml of the mixed solution was filtered and injected into a deoxygenated EPR tube. In situ EPR spectra were collected under visible light.

Data availability

Additional experimental details of characterization and catalysis, DFT calculations and DFT-calculated structures are available in the Supplementary Information and Supplementary Data.

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Author contributions

B.A., C.W., S.Y. and M.S. conceived the idea and designed the project. S.Y. and M.S. directed and supervised the research. B.A., Z.L., X.Z., Y.C. and Z.Z. performed the synthesis, catalysis and mechanism study. Z.W., A.M.S., Z.L., F.T. and E.J.L.M. contributed to the EPR experiments. Z.L. and Y.C. performed the computational work. X.H., M.D.F., L.S.N., A.J.R.-C., C.W. and W.L. contributed to the data analysis. B.A., Z.L., S.Y. and M.S. wrote the paper, with input from all authors.

Competing interests

The authors declare no competing interests.

Additional information

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